

Nama : Rengganis Tantri Pramudita

NIM : 2311104065

Kelas : SE0702

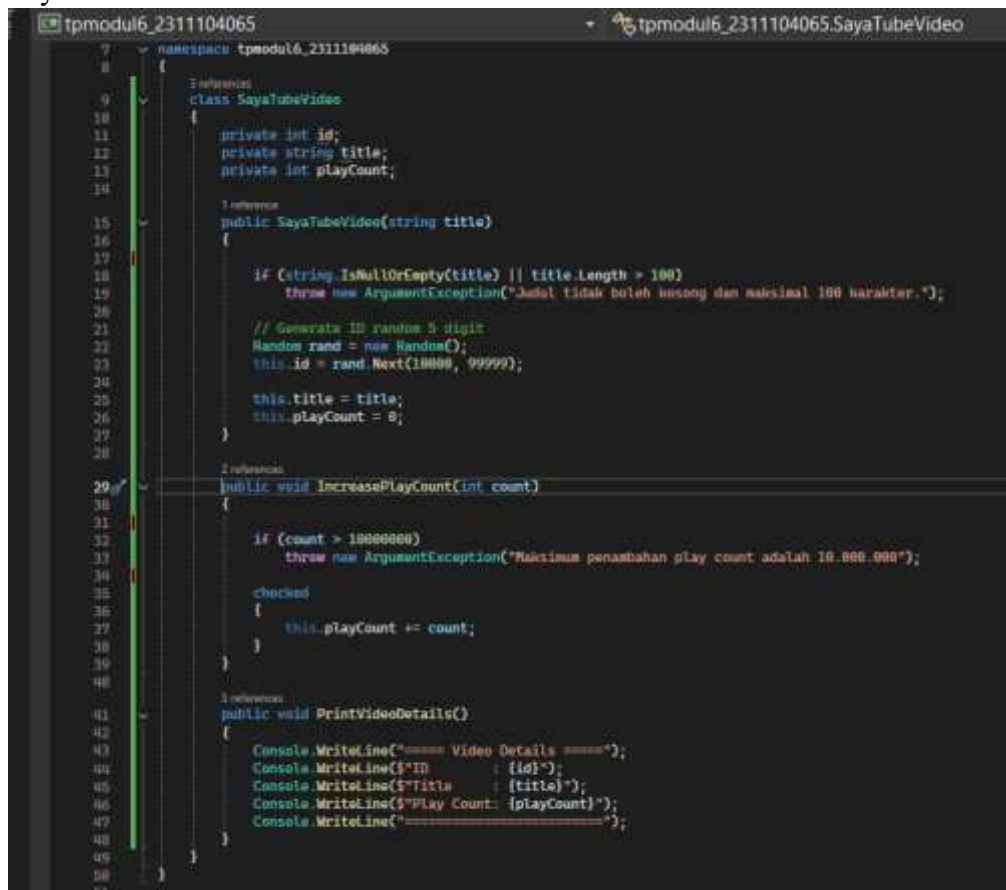
TP Modul 6

1. MEMBUAT PROJECT CONSOLE/TANPA GUI (Nama project tpmodul6_2311104065)
2. MELAKUKAN GIT COMMIT PADA PROJECT YANG DIBUAT

```
PS C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065> git init
Reinitialized existing Git repository in C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065\
PS C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065> git commit -m "initial proje
k"
[master (root-commit) 1a1a1a1] initial proje
k
1 file changed, 1 insertion(+), 0 deletions(-)
 create mode 100644 .gitignore
PS C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065> git branch -M main
PS C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065> git remote add origin https:
//github.com/rengganistan/tpmodul6_2311104065.git
error: remote origin already exists.
PS C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065> git remote remove origin
PS C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065> git remote add origin https:
//github.com/rengganistan/tpmodul6_2311104065.git
PS C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065> git push -u origin main
Enumerating objects: 34, done.
Counting objects: 100% (34/34), done.
Delta compression using up to 2 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (34/34), 34.79 MiB | 2.03 MiB/s, done.
Total 34 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/rengganistan/tpmodul6_2311104065.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'
PS C:\NPL_Rengganis Tantri Pramudita_2311104065_515E-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodul6_2311104065>
```

3. MENAMBAHKAN KODE AWAL SAYATUBEVIDEO

- SayaTubeVideo.cs



```
1 namespace tpmodul6_2311104065
2 {
3     3 references
4     class SayaTubeVideo
5     {
6         11 private int id;
7         12 private string title;
8         13 private int playCount;
9
10    14
11    15 1 reference
12    public SayaTubeVideo(string title)
13    {
14        18 if (string.IsNullOrEmpty(title) || title.Length > 100)
15        19     throw new ArgumentException("Judul tidak boleh kosong dan maksimal 100 karakter.");
16
17        21 // Generate ID random 5 digit
18        Random rand = new Random();
19        22 this.id = rand.Next(10000, 99999);
20
21        23 this.title = title;
22        24 this.playCount = 0;
23
24        25
25        26
26        27
27        28
28        29 2 references
29        public void IncreasePlayCount(int count)
30        {
31            32
32            33 if (count > 10000000)
33            34     throw new ArgumentException("Maksimum penambahan play count adalah 10.000.000");
34
35            35 checked
36            {
37                37 this.playCount += count;
38            }
39
40            40
40            41 1 reference
41            public void PrintVideoDetails()
42            {
43                43 Console.WriteLine("===== Video Details =====");
44                44 Console.WriteLine($"ID : {id}");
45                45 Console.WriteLine($"Title : {title}");
46                46 Console.WriteLine($"Play Count: {playCount}");
47                47 Console.WriteLine("=====");
48            }
49        }
50    }
51 }
```

```

namespace tpmodul6_2311104065
{
    class SayaTubeVideo
    {
        private int id;
        private string title;
        private int playCount;

        public SayaTubeVideo(string title)
        {
            if (string.IsNullOrEmpty(title) || title.Length > 100)
                throw new ArgumentException("Judul tidak boleh kosong dan maksimal 100 karakter.");

            // Generate ID random 5 digit
            Random rand = new Random();
            this.id = rand.Next(10000, 99999);

            this.title = title;
            this.playCount = 0;
        }

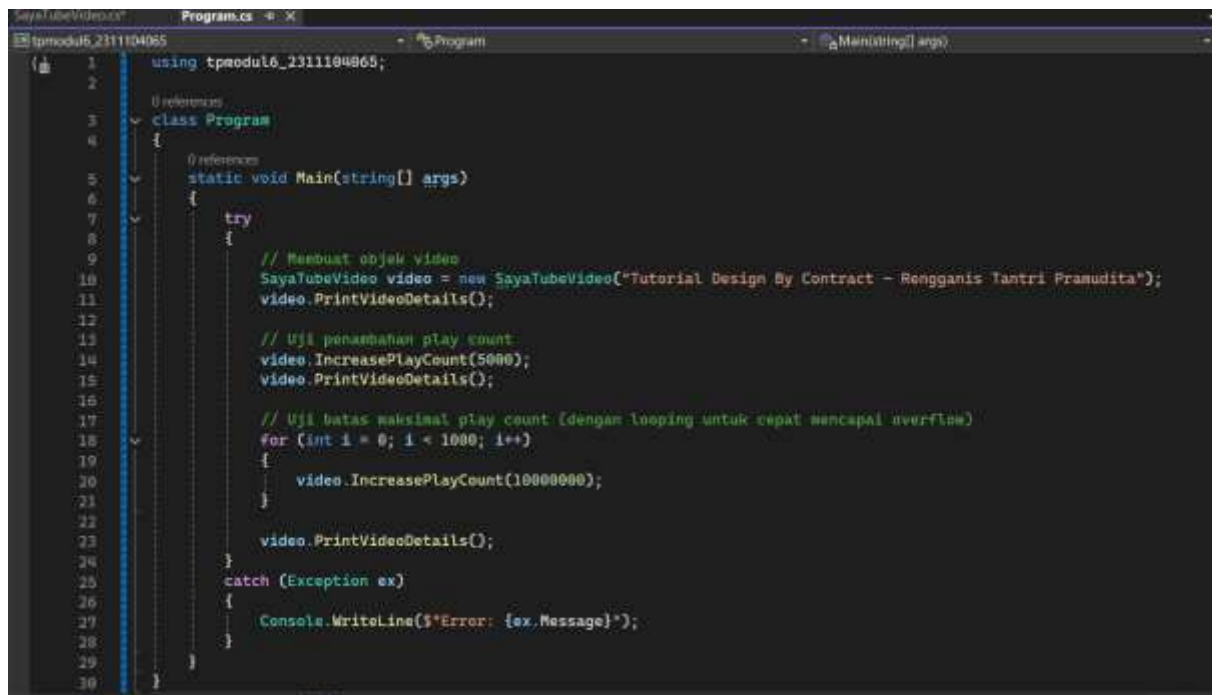
        public void IncreasePlayCount(int count)
        {
            if (count > 10000000)
                throw new ArgumentException("Maksimum penambahan play count adalah 10.000.000");

            checked
            {
                this.playCount += count;
            }
        }

        public void PrintVideoDetails()
        {
            Console.WriteLine("===== Video Details =====");
            Console.WriteLine($"ID      : {id}");
            Console.WriteLine($"Title   : {title}");
            Console.WriteLine($"Play Count: {playCount}");
            Console.WriteLine("=====");
        }
    }
}

```

- Program.cs



```
1 using tpmodul6_2311104065;
2
3 class Program
4 {
5     static void Main(string[] args)
6     {
7         try
8         {
9             // Membuat objek video
10            SayaTubeVideo video = new SayaTubeVideo("Tutorial Design By Contract - Rengganis Tantri Pramudita");
11            video.PrintVideoDetails();
12
13            // Uji penambahan play count
14            video.IncreasePlayCount(5000);
15            video.PrintVideoDetails();
16
17            // Uji batas maksimal play count (dengan looping untuk cepat mencapai overflow)
18            for (int i = 0; i < 1000; i++)
19            {
20                video.IncreasePlayCount(10000000);
21            }
22
23            video.PrintVideoDetails();
24        }
25        catch (Exception ex)
26        {
27            Console.WriteLine($"Error: {ex.Message}");
28        }
29    }
30 }
```

```
using tpmodul6_2311104065;

class Program
{
    static void Main(string[] args)
    {
        try
        {

            SayaTubeVideo video = new SayaTubeVideo("Tutorial Design By Contract – Rengganis
Tantri Pramudita");
            video.PrintVideoDetails();

            video.IncreasePlayCount(5000);
            video.PrintVideoDetails();

            for (int i = 0; i < 1000; i++)
            {
                video.IncreasePlayCount(10000000);
            }

            video.PrintVideoDetails();
        }
        catch (Exception ex)
        {
            Console.WriteLine($"Error: {ex.Message}");
        }
    }
}
```

- Hasil run:

```
Microsoft Visual Studio Debu x + v

===== Video Details =====
ID      : 64703
Title   : Tutorial Design By Contract - Rengganis Tantri Pramudita
Play Count: 0
=====
===== Video Details =====
ID      : 64703
Title   : Tutorial Design By Contract - Rengganis Tantri Pramudita
Play Count: 5000
=====
Error: Arithmetic operation resulted in an overflow.

C:\KPL_Rengganis Tantri Pramudita_2311104065_S1SE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodule6_2311104065.exe (process 12844) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automatically close console when debugging stops.
Press any key to close this window . . .
```

4. MELAKUKAN COMMIT KEDUA

```
PS C:\KPL_Rengganis Tantri Pramudita_2311104065_S1SE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodule6_2311104065> git add .
PS C:\KPL_Rengganis Tantri Pramudita_2311104065_S1SE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodule6_2311104065> git commit -m "menambahkan class SayaTubeVideo"
[main 3128b5f] menambahkan class SayaTubeVideo
33 files changed, 174 insertions(+), 0 deletions(-)
create mode 100644 .vs\tpmodule6_2311104065\FileContentIndex\fe339e90-f9ac-4b38-854d-fcb56af000e.vidx
create mode 100644 .vs\tpmodule6_2311104065\FileContentIndex\72d3f83f-b61f-4833-b396-b13615322611.vidx
create mode 100644 .vs\tpmodule6_2311104065\FileContentIndex\72ae890a-2ca8-4868-b86d-485437198c18.vidx
delete mode 100644 .vs\tpmodule6_2311104065\FileContentIndex\c9d7d88f-8859-47a2-a84c-07731572a834.vidx
create mode 100644 tpmodule6_2311104065\SayaTubeVideo.cs
create mode 100644 tpmodule6_2311104065\bin\Debug\net8.0\tpmodule6_2311104065.deps.json
create mode 100644 tpmodule6_2311104065\bin\Debug\net8.0\tpmodule6_2311104065.dll
create mode 100644 tpmodule6_2311104065\bin\Debug\net8.0\tpmodule6_2311104065.exe
create mode 100644 tpmodule6_2311104065\bin\Debug\net8.0\tpmodule6_2311104065.pdb
create mode 100644 tpmodule6_2311104065\bin\Debug\net8.0\tpmodule6_2311104065.runtimeconfig.json
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\apphost.exe
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\ref\tpmodule6_2311104065.dll
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\refint\tpmodule6_2311104065.dll
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\tpmodule6_2311104065.csproj.BuildWithSkipAnalyzers
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\tpmodule6_2311104065.csproj.CoreCompileInputs.cache
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\tpmodule6_2311104065.csproj.FileListAbsolute.txt
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\tpmodule6_2311104065.dll
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\tpmodule6_2311104065.genruntimeconfig.cache
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\tpmodule6_2311104065.pdb
create mode 100644 tpmodule6_2311104065\obj\Debug\net8.0\tpmodule6_2311104065.sourcelink.json
PS C:\KPL_Rengganis Tantri Pramudita_2311104065_S1SE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodule6_2311104065> git push -u origin main
fatal: unable to access 'https://github.com/rengganistan/tpmodule6_2311104065.git/': Failed to connect to github.com port 443 after 23118 ms: Couldn't connect to server
PS C:\KPL_Rengganis Tantri Pramudita_2311104065_S1SE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodule6_2311104065> git push origin main
Enumerating objects: 47, done.
Counting objects: 100% (46/46), done.
Delta compression using up to 2 threads
Compressing objects: 100% (39/39), done.
Writing objects: 100% (44/44), 95.34 KiB | 2.27 MiB/s, done.
Total 44 (delta 11), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (11/11), completed with 9 local objects.
To https://github.com/rengganistan/tpmodule6_2311104065.git
d2afde2..b32b5f: main -> main
PS C:\KPL_Rengganis Tantri Pramudita_2311104065_S1SE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpmodule6_2311104065>
```

5. MENAMBAHKAN IMPLEMENTASI DESIGN BY CONTRACT

- SayaTubeVideo.cs

```
SayaTubeVideo.cs* Program.cs
tpmodul6_2311104065 tpmodul6_2311104065.SayaTubeVideo

7 namespace tpmodul6_2311104065
8 {
9     public class SayaTubeVideo
10     {
11         private int id;
12         private string title;
13         private int playCount;
14
15         public SayaTubeVideo(string title)
16         {
17             if (string.IsNullOrEmpty(title) || title.Length > 100)
18                 throw new ArgumentException("Judul tidak boleh kosong dan maksimal 100 karakter.");
19
20             Random rand = new Random();
21             this.id = rand.Next(10000, 99999);
22
23             this.title = title;
24             this.playCount = 0;
25         }
26
27         public void IncreasePlayCount(int count)
28         {
29             if (count > 10000000)
30                 throw new ArgumentException("Maksimum penambahan play count adalah 10.000.000");
31
32             try
33             {
34                 checked
35                 {
36                     this.playCount += count;
37                 }
38             }
39             catch (OverflowException)
40             {
41                 Console.WriteLine("Error: Terjadi overflow saat menambah play count.");
42             }
43         }
44
45         public void PrintVideoDetails()
46         {
47             Console.WriteLine("==== Video Details =====");
48             Console.WriteLine($"ID : {id}");
49             Console.WriteLine($"Title : {title}");
50             Console.WriteLine($"Play Count: {playCount}");
51             Console.WriteLine("=====");
52         }
53     }
54 }
```

```
namespace tpmodul6_2311104065
{
    public class SayaTubeVideo
    {
        private int id;
        private string title;
        private int playCount;

        public SayaTubeVideo(string title)
        {
            if (string.IsNullOrEmpty(title) || title.Length > 100)
                throw new ArgumentException("Judul tidak boleh kosong dan maksimal 100 karakter.");

            Random rand = new Random();
            this.id = rand.Next(10000, 99999);

            this.title = title;
            this.playCount = 0;
        }

        public void IncreasePlayCount(int count)
        {
```

```

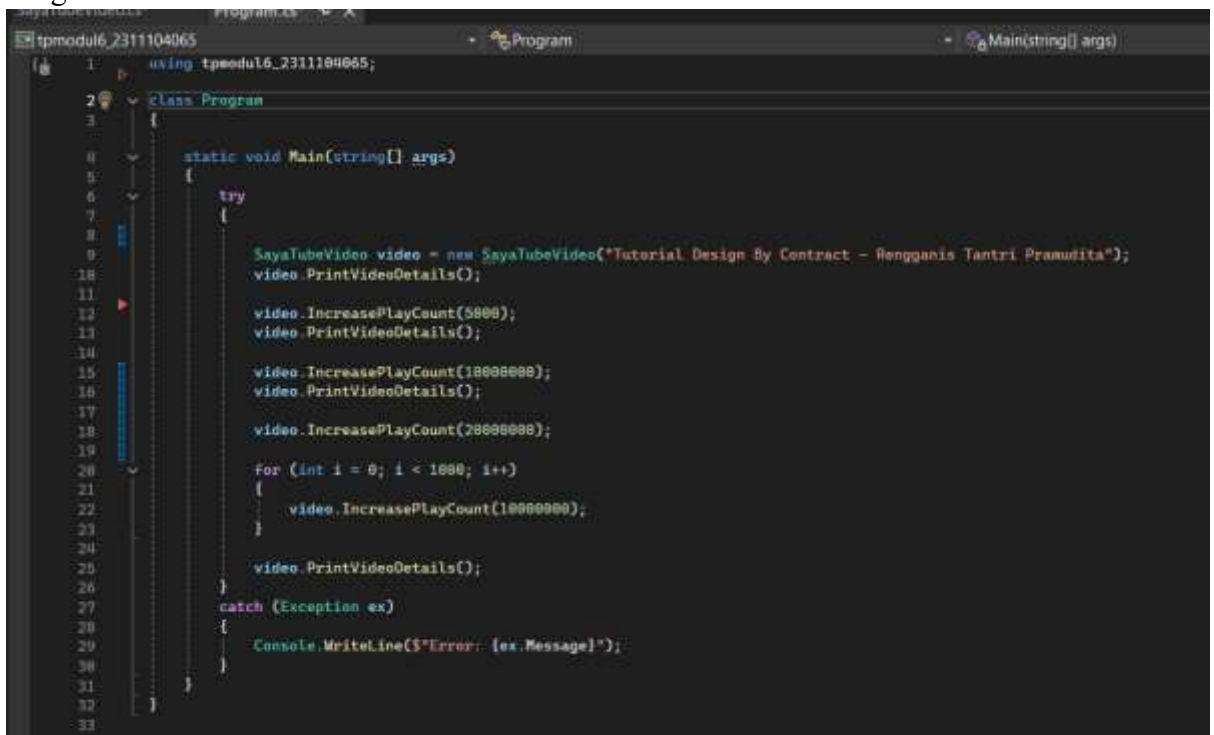
        if (count > 10000000)
            throw new ArgumentException("Maksimum penambahan play count adalah 10.000.000");

        try
        {
            checked
            {
                this.playCount += count;
            }
        }
        catch (OverflowException)
        {
            Console.WriteLine("Error: Terjadi overflow saat menambah play count.");
        }
    }
}

public void PrintVideoDetails()
{
    Console.WriteLine("==== Video Details =====");
    Console.WriteLine($"ID      : {id}");
    Console.WriteLine($"Title   : {title}");
    Console.WriteLine($"Play Count: {playCount}");
    Console.WriteLine("=====");
}
}
}

```

- Program.cs



```

1  using tpmodule6_2311104065;
2
3  class Program
4  {
5
6      static void Main(string[] args)
7      {
8          try
9          {
10             SayaTubeVideo video = new SayaTubeVideo("Tutorial Design By Contract - Rengganis Tantri Pramudita");
11             video.PrintVideoDetails();
12
13             video.IncreasePlayCount(5000);
14             video.PrintVideoDetails();
15
16             video.IncreasePlayCount(10000000);
17             video.PrintVideoDetails();
18
19             video.IncreasePlayCount(20000000);
20
21             for (int i = 0; i < 1000; i++)
22             {
23                 video.IncreasePlayCount(10000000);
24             }
25
26             video.PrintVideoDetails();
27         }
28         catch (Exception ex)
29         {
30             Console.WriteLine($"Error: {ex.Message}");
31         }
32     }
33 }

```

```

using tpmodule6_2311104065;
class Program
{
    static void Main(string[] args)
    {
        try
        {

```

```

        SayaTubeVideo video = new SayaTubeVideo("Tutorial Design By Contract – Rengganis
Tantri Pramudita");
        video.PrintVideoDetails();

        video.IncreasePlayCount(5000);
        video.PrintVideoDetails();

        video.IncreasePlayCount(10000000);
        video.PrintVideoDetails();

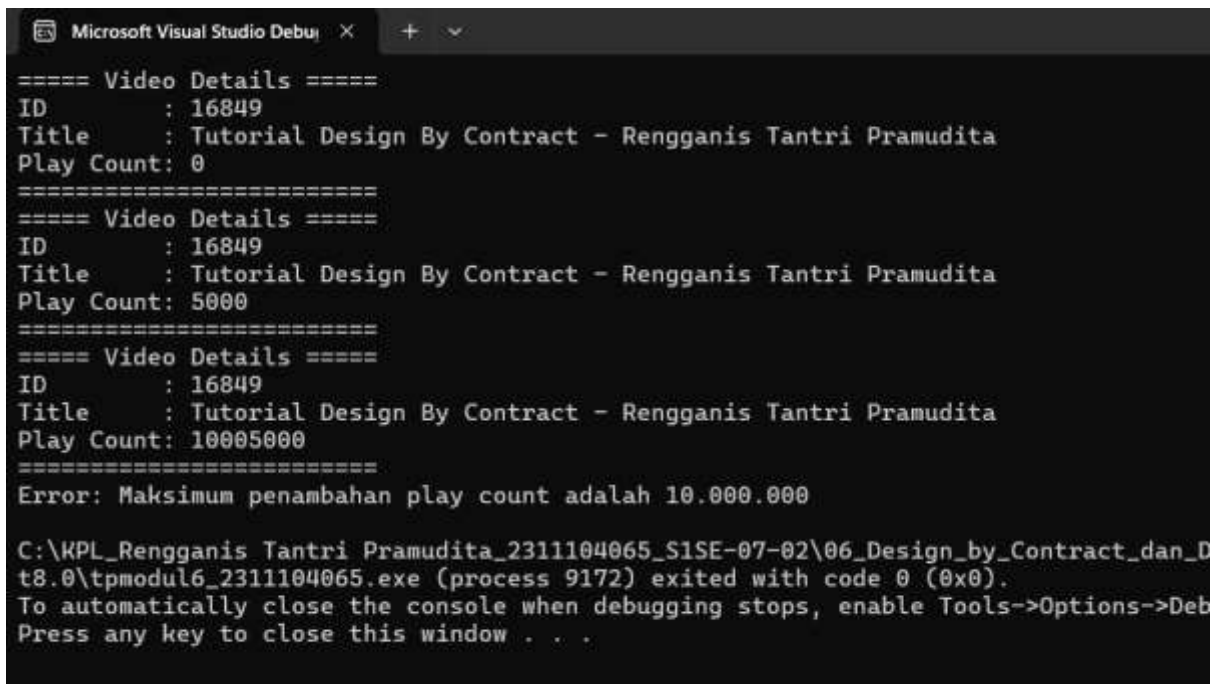
        video.IncreasePlayCount(20000000);

        for (int i = 0; i < 1000; i++)
        {
            video.IncreasePlayCount(10000000);
        }

        video.PrintVideoDetails();
    }
    catch (Exception ex)
    {
        Console.WriteLine($"Error: {ex.Message}");
    }
}
}

```

- Hasil run:



```

Microsoft Visual Studio Debug Console
===== Video Details =====
ID       : 16849
Title    : Tutorial Design By Contract - Rengganis Tantri Pramudita
Play Count: 0
=====
===== Video Details =====
ID       : 16849
Title    : Tutorial Design By Contract - Rengganis Tantri Pramudita
Play Count: 5000
=====
===== Video Details =====
ID       : 16849
Title    : Tutorial Design By Contract - Rengganis Tantri Pramudita
Play Count: 10005000
=====
Error: Maksimum penambahan play count adalah 10.000.000

C:\KPL_Rengganis Tantri Pramudita_2311104065_SISE-07-02\06_Design_by_Contract_dan_D
t8.0\tpmodul6_2311104065.exe (process 9172) exited with code 0 (0x0).
To automatically close the console when debugging stops, enable Tools->Options->Deb
Press any key to close this window . . .

```

6. MELAKUKAN COMMIT TERAKHIR

```

PS C:\WPL_Rengganis Tantri Pramudita_23111804865_SISE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpomodul6_23111804865> git add .
error: open('vs/tpomodul6_23111804865/FileContentIndex/80c144b1-3c13-4051-9907-caa5aa08056.vidx'): Permission denied
error: unable to index file 'vs/tpomodul6_23111804865/FileContentIndex/80c144b1-3c13-4051-9907-caa5aa08056.vidx'
fatal: adding files failed
PS C:\WPL_Rengganis Tantri Pramudita_23111804865_SISE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpomodul6_23111804865> git add .
PS C:\WPL_Rengganis Tantri Pramudita_23111804865_SISE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpomodul6_23111804865> git commit -s "Implementasi
design by contract"
[main 1d984b7] Implementasi design by contract
21 files changed, 36 insertions(+), 30 deletions(-)
delete mode 1806400 vs/tpomodul6_23111804865/FileContentIndex/7e539e98-f9ec-4b38-954d-46cb56af940e.vidx
delete mode 1806400 vs/tpomodul6_23111804865/FileContentIndex/7243f5a9-861f-4533-b396-b15619322611.vidx
create mode 1806400 vs/tpomodul6_23111804865/FileContentIndex/8d9bd885-ab3a-49de-8bbd-dc529cdad7f8.vidx
create mode 1806400 vs/tpomodul6_23111804865/FileContentIndex/c007cffe-c571-4f2d-a028-a417097393aa.vidx
PS C:\WPL_Rengganis Tantri Pramudita_23111804865_SISE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpomodul6_23111804865> git push origin main
Enumerating objects: 23, done.
Counting objects: 100% (53/53), done.
Delta compression using up to 3 threads
Compressing objects: 100% (26/26), done.
Writing objects: 100% (20/20), 10.10 KiB | 1.13 MiB/s, done.
Total 20 (delta 16), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (16/16), completed with 15 local objects.
To https://github.com/rengganistan/tpomodul6_23111804865.git
 d32808f..1d984b7 main -> main
PS C:\WPL_Rengganis Tantri Pramudita_23111804865_SISE-07-02\06_Design_by_Contract_dan_Defensive_Programming\tpomodul6_23111804865>

```